

OpenHPLjl HydraulicPort

potential: $H(t)$ [m]

flow: $Q(t)$ [m³/s]

$Q > 0$: into component

MTK: Q [connect = Flow]

HydroPowerDynamics.jl HydraulicPort

potential: $p(t)$ [Pa]

flow: $\dot{m}(t)$ [kg/s]

$\dot{m} > 0$: into component

MTK: dm [connect = Flow]

same conservation idea

Physical conversion

$$p = \rho g H + p_{\text{ref}}$$

$$\dot{m} = \rho Q$$

Rotational convention

across: ω (and ϕ in HPD)

flow: τ

$\tau > 0$: into component